

Subject: Programming with Java

ASSIGNMENT No :2

TITLE :Develop a Java application demonstrating the use of default, parameterized, and copy constructors for object initialization and data management.

Assignment 2: Methods and Static Members

Problem Statement

Develop a Java application demonstrating the use of default, parameterized, and copy constructors for object initialization and data management.

Learning Objectives

To understand and implement default, parameterized, and copy constructors for object initialization.

Theory

Constructors are special member functions that are automatically invoked when an object is created. Java supports default constructors provided by the compiler, parameterized constructors for initializing objects with user-defined values, and copy constructors (implemented manually) for creating duplicate objects. Constructors improve code readability and simplify object initialization while supporting object-oriented programming principles.

Conclusion

Hence, the implementation of various constructor types in Java was successfully demonstrated to handle object initialization and data management effectively. By utilizing default constructors for baseline setup, parameterized constructors for assigning specific user-defined values, and copy constructors for creating accurate duplicate records, the exercises illustrated flexible and efficient object creation. This approach not only improves overall code readability and simplifies the initialization process, but it also strongly reinforces core object-oriented programming principles by ensuring data consistency and reusability across multiple instances.

Exercise

1. Create a Student class using default and parameterized constructors to initialize student name and roll number.

Code:

```
1 public class Student {  
2     String name;  
3     long prn;  
4  
5     public Student() {  
6         name = "Not Assigned";  
7         prn = 0L;  
8     }  
9  
10    public Student(String n, long p) {  
11        name = n;  
12        prn = p;  
13    }  
14  
15    public void display() {  
16        System.out.println("Name: " + name + ", PRN: " + prn);  
17    }  
18  
19    public static void main(String[] args) {  
20        Student s1 = new Student();  
21        Student s2 = new Student("Anurag Thakur", 25070122040L);  
22  
23        s1.display();  
24        s2.display();  
25    }  
26 }
```

Output:

```
Name: Not Assigned, PRN: 0  
Name: Anurag Thakur, PRN: 25070122040  
  
=== Code Execution Successful ===
```

2. Develop a Mobile Phone Inventory System using different constructors to initialize mobile details and create duplicate object records.

Code:

```
1  public class MobilePhone {
2      String model;
3      int price;
4
5      public MobilePhone() {
6          model = "Unknown";
7          price = 0;
8      }
9
10     public MobilePhone(String m, int p) {
11         model = m;
12         price = p;
13     }
14
15     public MobilePhone(MobilePhone mobile) {
16         model = mobile.model;
17         price = mobile.price;
18     }
19
20     public void show() {
21         System.out.println("Model: " + model + ", Price: $" + price);
22     }
23
24     Run main | Debug main
25     public static void main(String[] args) {
26         MobilePhone m1 = new MobilePhone("Galaxy S24", 800);
27
28         MobilePhone m2 = new MobilePhone(m1);
29
30         m1.show();
31         m2.show();
32     }
```

Output:

```
Listening on 59535
User program running
② Model: Galaxy S24, Price: $800
User program finished
```

